

DP-LLM: Problem Setting and Research Gap

Goal: adapt one on-device LLM to changing latency, memory, and quality constraints.

Past research: static PTQ

Weight-only PTQ cuts memory traffic.
Precision is fixed after calibration.
Cannot react to per-query runtime budgets.

Past research: multi-scale

Multiple bit-width views share storage.
Layer-wise mixed precision supports non-integer bits.
Assignments remain offline or static during decoding.

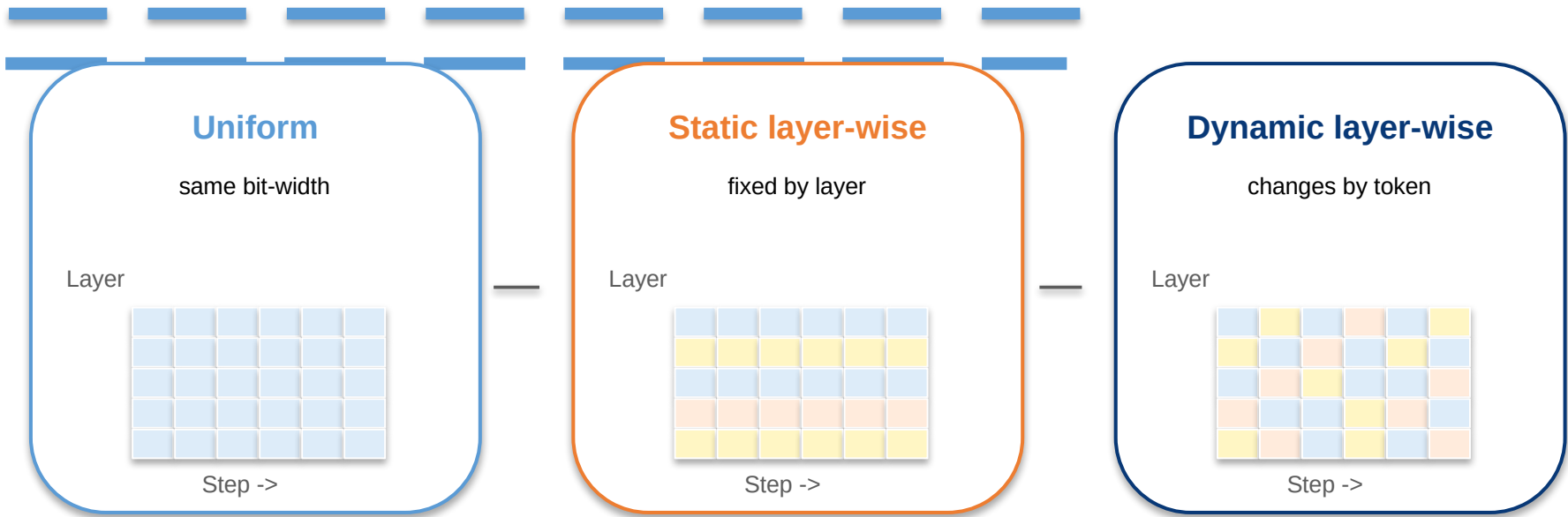
What is missing

How to match target precision or latency at runtime.
Layer sensitivity changes token by token.
Static precision leaves trade-offs unused.

DP-LLM treats precision as a runtime, layer-wise decision

Core Observation: Sensitivity Changes by Token

The high-precision layers are not fixed; sensitivity shifts across decoding steps.



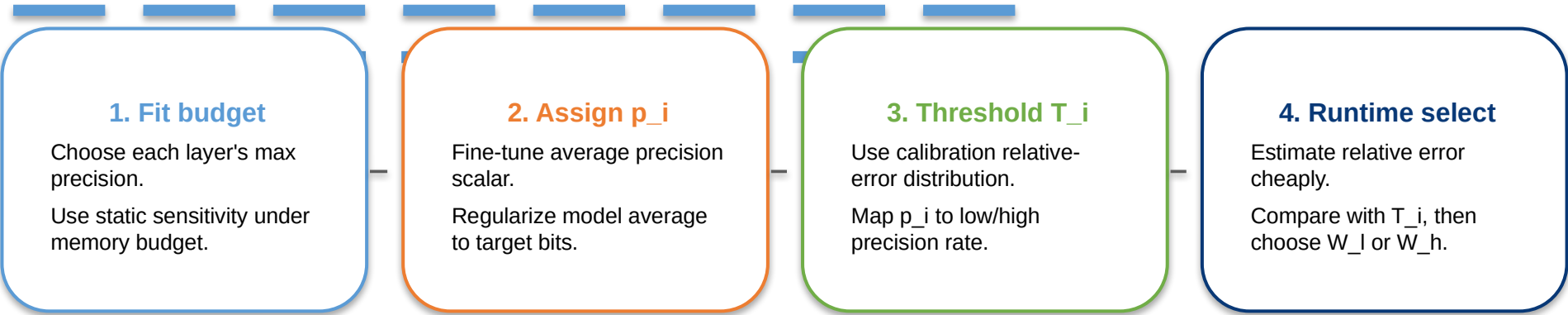
Key observation: dynamic layer-wise mixed precision captures token-step sensitivity

The paper uses an oracle analysis to show that dynamic assignment can lower perplexity versus static layer-wise mixed precision.

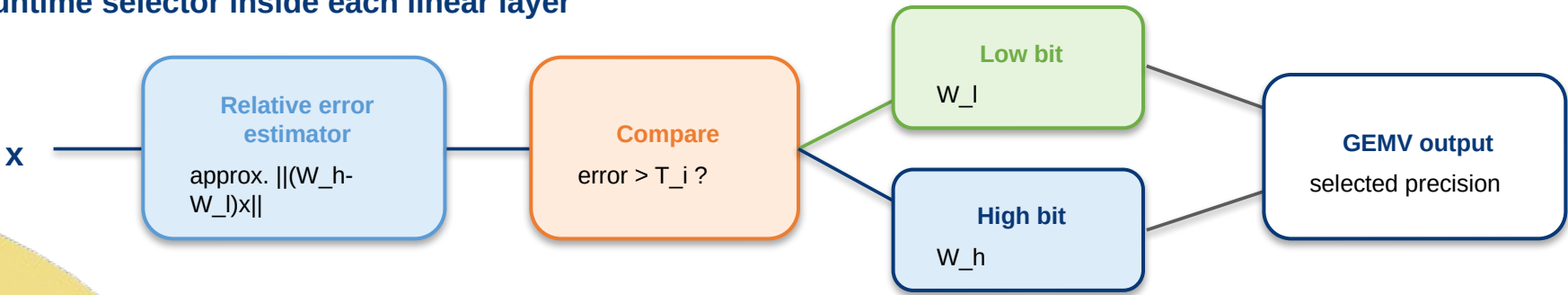
Source: paper Figures 2-3; Section 2.4

Method: DP-LLM Precision Selector

Offline thresholds plus runtime selectors choose low or high precision per layer.



Runtime selector inside each linear layer



Efficiency trick: use linear-regression estimation when $\|x\|$ predicts error; otherwise use random projection. Async estimation hides part of the cost.

Source: paper Sections 3-5; Figures 4-6

Main Results and Takeaways

DP-LLM improves quality versus static mixed precision with small runtime overhead.

Quality

Models: Llama-3-8B and Phi-3-Medium.
Lower PPL than LLM-MQ / HAWQ-V2 on WikiText2 and C4 in most settings.
Often strongest on GSM8K, MBPP, BBH, and MATH.

Runtime overhead

Selector overhead geomean:
1.45% for Llama-3-8B; 0.81% for Phi-3-Medium.
RTX 4060Ti Hybrid+Async: 0.74%, 0.66%, 0.45% at 3.5/4.0/4.5 bits.

Conclusion

Relative error works as a precision-choice proxy.
Dynamic assignment captures sensitivity missed by static baselines.
99th percentile effective-bit drift is about 2.25-3.32%.

Takeaway: target precision becomes low-overhead per-layer runtime adaptation

Benchmarks: WikiText2, C4, GSM8K, MBPP, BBH, MATH; hardware: Jetson Orin AGX and RTX 4060 Ti.

Source: paper Tables 1-7; Section 6